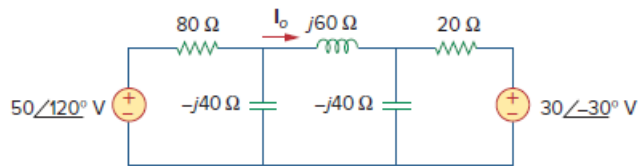


Problem

Use mesh analysis to determine current I_o in the circuit of Fig. 10.79 below.

Figure 10.79:



Step-by-step solution

Step 1 of 7

Refer to Figure 10.79 in the textbook.

Re-draw the circuit diagram with mesh currents.

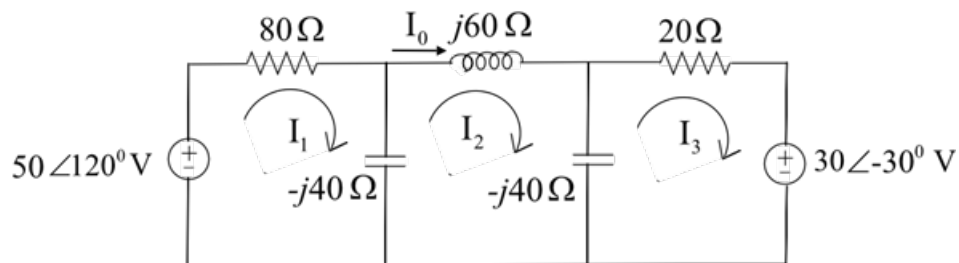


Figure 1

Step 2 of 7

Apply Kirchhoff's voltage law to mesh 1.

$$80I_1 + (-j40)(I_1 - I_2) - 50\angle 120^\circ = 0$$

$$I_1(80 - j40) - I_2(-j40) - 50\angle 120^\circ = 0$$

$$I_1(8 - j4) + I_2(j4) = 5\angle 120^\circ \quad \dots\dots (1)$$

Step 3 of 7

Apply Kirchhoff's voltage law to mesh 2.

$$(I_2 - I_1)(-j40) + j60I_2 + (I_2 - I_3)(-j40) = 0$$

$$I_1(j40) + I_2(j60 - j40 - j40) + I_3(j40) = 0$$

$$I_1(j40) + I_2(-j20) + I_3(j40) = 0$$

$$2I_1 - I_2 + 2I_3 = 0 \quad \dots\dots (2)$$

Step 4 of 7

Apply Kirchhoff's voltage law to mesh 3.

$$20I_3 + (-j40)(I_3 - I_2) + 30\angle -30^\circ = 0$$

$$I_2(j40) + I_3(20 - j40) = -30\angle -30^\circ$$

$$I_2(j4) + I_3(2 - j4) = -3\angle -30^\circ$$

$$\begin{aligned} I_3 &= \frac{-3\angle -30^\circ}{2 - j4} - \left(\frac{4\angle 90^\circ}{2 - j4} \right) I_2 \\ &= \frac{-3\angle -30^\circ}{4.47\angle -63.43^\circ} - \left(\frac{4\angle 90^\circ}{4.47\angle -63.43^\circ} \right) I_2 \\ &= -0.67\angle 33.43^\circ - (0.895\angle 153.43^\circ) I_2 \end{aligned}$$

$$I_3 = -0.67\angle 33.43^\circ + (0.8 - j0.4)I_2 \dots\dots (3)$$

Step 5 of 7

Substitute equation (3) in equation (2).

$$2I_1 - I_2 + 2(-0.67\angle 33.43^\circ + (0.8 - j0.4)I_2) = 0$$

$$2I_1 + (0.6 - j0.8)I_2 = 1.34\angle 33.43^\circ \dots\dots (4)$$

Write equation (1) and equation (4) in matrix form.

$$\begin{pmatrix} 8 - j4 & j4 \\ 2 & 0.6 - j0.8 \end{pmatrix} \begin{pmatrix} I_1 \\ I_2 \end{pmatrix} = \begin{pmatrix} 5\angle 120^\circ \\ 1.34\angle 33.43^\circ \end{pmatrix}$$

Step 6 of 7

Calculate the value of Determinant.

$$\begin{aligned} \Delta &= \begin{vmatrix} 8 - j4 & j4 \\ 2 & 0.6 - j0.8 \end{vmatrix} \\ &= (8 - j4)(0.6 - j0.8) - (2)(j4) \\ &= 1.6 - j16.8 \end{aligned}$$

Determine the value of Δ_2 .

$$\begin{aligned} \Delta_2 &= \begin{vmatrix} 8 - j4 & 5\angle 120^\circ \\ 2 & 1.34\angle 33.43^\circ \end{vmatrix} \\ &= (8 - j4)(1.34\angle 33.43^\circ) - (2)(5\angle 120^\circ) \\ &= (8.94\angle -26.57^\circ)(1.34\angle 33.43^\circ) - (2)(5\angle 120^\circ) \\ &= 11.98\angle 6.86^\circ - 10\angle 120^\circ \\ &= 16.89 - j7.23 \end{aligned}$$

Step 7 of 7

Apply Cramer's rule, determine the current in mesh 2.

$$\begin{aligned} I_2 &= \frac{\Delta_2}{\Delta} \\ &= \frac{16.89 - j7.23}{1.6 - j16.8} \\ &= \frac{18.37 \angle -23.17^\circ}{16.88 \angle -84.56^\circ} \\ &= 1.089 \angle 61.4^\circ \text{ A} \end{aligned}$$

From figure 1, $I_0 = I_2$.

Determine the value of I_0 .

$$\begin{aligned} I_0 &= I_2 \\ &= 1.088 \angle 61.4^\circ \text{ A} \end{aligned}$$

Therefore, the value of current, I_0 is $1.089 \angle 61.4^\circ \text{ A}$.